II. THE DESIGN AND IMPLEMENTATION OF ENVIRONMENTAL POLICY

I. CRITERIA FOR EVALUATING ENVIRONMENTAL POLICIES

There are many environmental policies in practice all over the world. Each policy has its own different characteristics, advantages and disadvantages. Each policy has certain expectations about the behavior of the administrators and the polluters. These policies however succeed in some aspects and failed in other aspects. Following criteria should be considered while framing an effective environmental policy.

1. Efficiency: Administration in implementing various policies is said to be efficient when it gives maximum benefits to the society or when it adds into the well-being of the society compare to the social cost. In case of environmental policies related to pollution control, it implies to maintain a balance between abatement costs and damages. It brings us to the equilibrium point where marginal abatement costs and marginal damages are equal.

Environmental policies can be Centralized and Decentralized. In a centralized policy a central agency or the regulatory agency is in charge to attain efficiency. It takes such steps to move to the point where marginal abatement costs equals the marginal damages. It is possible only when the regulatory agency is having full knowledge of the marginal abatement costs and marginal damages. In a decentralized policy approach, the interactions of many individual decision makers reveal the information about the marginal abatement costs and marginal damages. The assessment of these individual decision makers adjust to the situation towards the point where marginal abatement costs equals the marginal damages.

2. Cost effectiveness: Cost effectiveness criterion of environmental policy implies that the cost incurred to achieve environmental improvement should bring about a greater amount of environmental improvement at the least possible cost. In simple words the costs incurred for environmental improvement should be less than the benefits enjoyed by the society due to environmental improvement. The resources employed to environmental quality improvement must be used in such a manner so that it will have a huge impact. It is especially important in LDCs or Developing Countries where resources are limited and therefore these countries cannot afford to implement the policies which are not cost effective.

3. Fairness: Fairness criterion for evaluating environmental policies is justifiable on the principle of equity. There should be equal distribution of benefits among the members of the society. It is explained with the help of following example.Suppose there are several alternative air-pollution control policies available to the government for a particular region. The

				Distribution of Net benefits	
Air pollution control program	Total cost	Total benefits	Net benefits	Region X	Region Y
A	50	100	50	25	25
B	50	100	50	30	20
C	50	140	90	20	70
D	50	140	90	40	50

implementation of one such policy is depend upon the evaluation of cost and benefits of these alternatives.

The above table shows the four alternative air pollution control programs that can be adopted by the government i.e. A, B, C, and D. there are two regions X and Y where region A represents lower income group people and region Y represents high income group people. It is seen from the above table that net benefits of Program A and B are same. But the distribution of net benefit in region X is more than region Y i.e. distribution in low-income region is more. So, we may prefer Program B which justifies the principle of equity or fairness. In case of Program B and C again we may prefer B because the distribution of net benefits is more towards region Y representing high income group and therefore it is unequal or unfair. In case of Program B and D, we can see that net benefits are more in both the regions X and Y as compare to program B. So, we prefer D to B since region X is better of compare to the benefits it is getting if program B is adopted.

4. Incentives for technological improvements: The decision of the private parties, consumers actually determine the extent of environmental impact. If the incentives are given to these parties, then it will reduce the environmental damage extensively. So, whether the policy is providing incentive to adopt technological improvements to reduce environmental impact is one of the important criteria. It is inevitable to invent and innovate new technology which helps to reduce the environmental impact in the long run. Technological improvements or innovations through R & D programs shifts the marginal abatement cost function downward.

The invention and innovation of new technology is a long term and complex process. It requires that resources to be devoted to invention and development in the first place. Then the issues of copyright, patent and related infrastructure arise. Since the huge amount of monetary investment and time is required, the public sector has to take initiative to develop R & D in this area. Incentive to adopt new technology will give boost to the R & D process of new technology.

5. Enforceability: Mere making a law to restrict or regulate something is not enough. Policy makers have to take regulatory measures to enforce it. In case of many stringent laws people do not accept or obey them voluntarily. So, enforceability of law is inevitable. Therefore, enforceability is an important criterion for effective environmental policy. There are two main steps in enforceability of a policy

i) Monitoring: In monitoring, the performance of the polluters is measured in accordance with the relevant laws. The main objective of monitoring is to make polluters aware that they are abide by the law. Monitoring of pollution behaviour is a complicated procedure. Polluters try to evade this monitoring by adopting many ways. Technology can help to ease the process of monitoring.

ii) Sanctioning: In simple words sanctioning means penalties or fine imposed for violation of law. When monitoring process noticed that a polluter is violating a prescribed law, the policy makers refer his case to the court of law for levy of penalty or law. It may cause a burden on the judiciary system which is often costly and time consuming.

6. Moral consideration: Ethical consideration or moral consideration is another criterion for evaluating environmental policies. Different people have different kind of opinion to look at a policy. But a person's ethical approach or moral consideration helps to decide right or wrong choices for the various environmental policies. Welfare of majority should be given the priority. For e.g., if industrially developed countries are responsible for more environmental pollution is a fact, then they should bear a major burden in rectifying the environmental degradation situation is a general opinion.

In this way, these six types of criteria are useful in evaluating environmental policies in different circumstances.

II. ENVIRONMENTAL STANDARDS

In simpler terms environmental standards refer to the maximum permissible concentration of harmful or potentially harmful pollutants which can be released that guarantees an acceptable health condition or environmental quality. Environmental standards help in setting a permissible limit of the pollutants to protect the human and environment health.

In other words, these standards are numerical values and range of permissible limit prescribed by the governments across the globe and implemented by regulatory agencies in the respective countries to decide the environmental quality. Therefore, it is obvious that it may vary from one nation to other nation and geographical location. The obvious question which might be striking to you is that why do we set such limits? Simply, this is because it helps to control the concentration of pollutants, effluents waste or industrial discharge to land, water and air.

This is generally expressed in terms of the permissible concentration of pollutants in unit of air emitted or in waste water discharge from source. It can also be expressed in terms of total load of pollutants per time unit, unit of production or unit of energy or materials inputs.

There are three types of environmental standards:

1. Ambient Standards
2. Emission Standards
3. Technology Standards

1. Ambient Standards: An ambient environmental standard refers to the surrounding concentration of pollutants which cannot be explored directly based upon the various emissions that lead to ambient quality level. It is very important to know the emission source of pollutants to achieve ambient standards. Sources *of pollutants may be helpful to achieve the standards.* Ambient standards expressed in terms of average concentration level or pollutants generated from various point and non-point sources.

Meteorological and hydrological phenomena play an important role in ambient standards. That is the reason for taking an average over a time period or over a geographical area. National Ambient Air Quality Standards set by Central Pollution Control Board is based on 24 hours and 8 hours.

Emission Standards: Emission standards are the permissible limits which never-exceeds beyond set limit applied directly coming from an industry or other sources. Emission standards are based on data related to the emission generation performance and cost of available technology.

The emission or effluent standards is related to the preferred method of control. The great advantage of such standards is their direct relevance to the polluting activity. It can be set on a wide variety of different bases.

For examples,

i) emission rate (e.g., kilograms per hour),

ii) emission concentration (e.g., parts per million of biochemical oxygen demand, or BOD, in waste-water),

iii) total quantity of residuals,

iv) residuals produced per unit of output (e.g., SO2 emissions per kilo watt hour of electricity produced, grams of CO per tonne of asphalt produced),

v) residual content per unit of input (e.g., sulphur content of coal used in power generation),

vi) Percentage removal of pollutants (e.g., 60-per-cent removal of waste materials before discharge).

Technology Standards: Technology standards specify the technologies, techniques, or practices including design, engineering, input and output standards which polluters must adopt or meet to protect the environment during manufacturing stage. In contrast to emission standards, technology standards impose on polluters certain decisions and technologies to be used. This is some form of "technology forcing' for polluting industries to adopt technological change in order to meet environmental standards.

III. PIGOVIAN TAXES

The British economist Arthur C. Pigou developed the concept of externality in early 1900. He advocated that Pigouvian tax is a tax imposed on those economic activities which generate negative externalities, which are borne by the parties not related to it. The market becomes inefficient since the costs which arises from negative externalities are not reflected in the final cost of a product or service. The main objective of Pigouvian tax to remove market inefficiencies by increasing the marginal private cost by the amount generated by the negative externality. In that case the final cost i.e. original cost-plus tax will reflect the full social cost of the economic activity. So the main purpose of Pigouvian tax is to discourage those activities which imposes a cost of production on to the unrelated persons in the society such as environmental pollution. This externality or external damage need to be corrected.

Pigouvian tax is imposed by the government on any activity which creates socially harmful externality. It is intended to tax the producer of the products or services who are responsible to create adverse side effects for society. And the amount of this tax is generally equivalent to the harm it causes to the society in the form of negative externality.

Pigouvian taxes are prevalent in the society in the form of tax on carbon emission, use of plastic bags etc. which creates negative externality. The implemented taxes are a measure to redistribute those costs back to the producer and/or user that generate the negative externality.

At Point A the market is efficient, the supply meets demand.

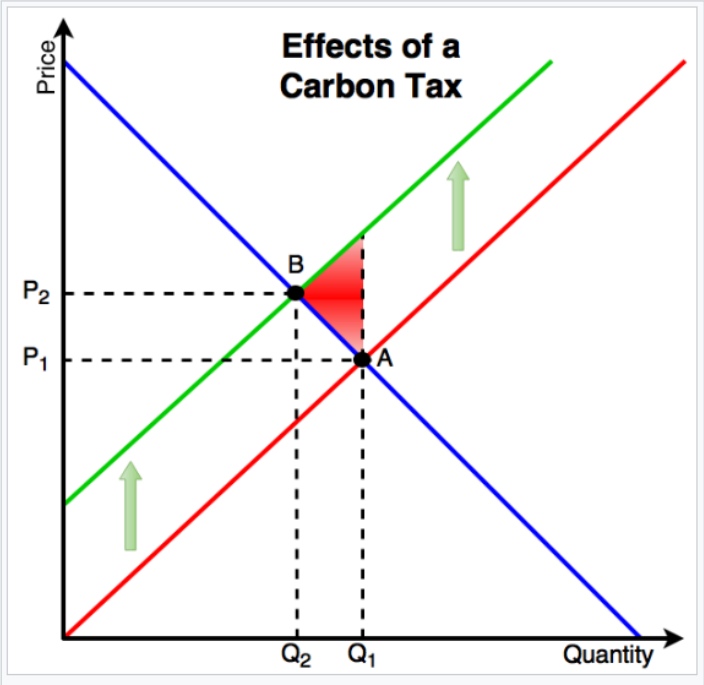
The market is not socially efficient however because the cost to others is not accounted for.

To compensate for the negative effects of an externality, a tax is imposed, raising the price from P1 to P2.

The implementation on the tax and the subsequent rise in the price to the consumer reduces the quantity demanded from Q1 to Q2.

The tax reduces the demand for electricity, reducing the overall emissions and the revenue generated can be used to remediate environmental damage or invest in cleaner generation technologies.

At Point B the market is socially efficient as the external costs to others have been internalized using the tax.



IV EFFLUENT TAX

A fee or tax that is paid by a polluter in exchange for the right to discharge an agreed amount of noxious emissions into the air and water. As a country's economy develops rapidly with rapidly expanding industrial economy, the demand for the use of water increases. Discharge of effluents or industrial waste by these growing industries into the water bodies is the main cause of water pollution. Water being a natural resource is scarce if not used effectively. Water pollution adds to this problem of scarcity furthermore. So effective water management is required which takes care of not only the quantity but also the quality of available water.

To justify the principle of polluter pays, effluent charges or fees are an instrument of pollution control, imposing a financial or economic costs on the polluter or producer who discharges effluents or wastes into the water bodies causing water pollution. These charges or fees are also called as pollution charges. The main objective of these fees is to manipulate the economic behaviour of the polluters. They can reduce the burden of these fees by

- By ceasing the production
- By bringing change in the technology of their production units which causes less pollution.

The amount collected by these effluent fees can be used by the government to improve the quality of water. The charges would be limited in total amount to a sum (and that sum would be earmarked in advance) sufficient to build and maintain those treatment facilities which in each particular receiving water body are necessary to achieve the quality of water inherent in the legislatively established classification for that particular water body.

Two different types of incentives would be given to the producers. These units can scrutinize their waste disposal program carefully and make any technological change which would reduce that charge by an amount greater than the cost of the change. Secondly in the long run, there will be a locational incentive. The units which need any quality water and the units whose

discharge of effluent is large will tend to locate nearby to those stream basins with low water quality standards. Whereas for the units which require high quality water or whose effluent discharge quantity is small will be located to stream basins having a high water quality standard.

To conclude, following elements should be considered for a desirable effluent fee:

- Effluent fees should justify and reflect the environmental cost of water pollution caused by effluents
- Effluent fees should bear some relation to Marginal abatement cost borne by the producer so that the producer will be motivated to reduce pollution
- Enough revenue should be generated for the maintenance and treatment of water bodies
- Enough economic incentive in the form of offsetting of pollution charges to be given to the producer who releases clean effluent into the water bodies.

V.TRADABLE PERMITS

Tradable permits are used in an environmental regulatory scheme where the sources of the pollutant to be regulated are given permits to release a specified number of tons of the pollutant. Tradable permits aim to reduce pollution emissions. There are two ways to issue these permits;

i) To auction it to the highest bidder; or

ii) To give it to the existing polluters in proportion to their baseline emissions.

The permit owners then free to buy or sell to anybody i.e. to other polluters who wants to participate in this market.

The tradable pollution permits are also called as cap and trade scheme.

This system offers flexibility to the polluters to attain the environmental goal. It offers a legal right to the companies to pollute or to emit a certain amount during a fixed time span. A firm who pollutes less can sell its leftover permit to pollute to another firm whose emission is more. Selling of its right to pollute to another firm provides an incentive to develop innovative solutions.

The basic purpose of the tradable permit is to raise revenue by selling the permits and to reduce pollution over a period of time. If a firm fails to abide by the permissible limit then it has to buy this from other firms or from the government. The price of this permit in the market is therefore determine by the demand for and supply of these permits. In the long run the government can reduce the size of tradable permits and by selling it at a higher rate can induce the firms to innovate or adopt the measures which reduces the pollution or emission and thereby to reduce the cost of this pollution. The overall impact will be of reducing of the pollution by the firms.

When a firm creates or emits carbon during its production, then it creates negative externality. In this case social marginal cost of production is more than the private marginal cost. Tradable permits are a method to try and reduce output to a more socially desirable level. The price of the tradable permits is set to cover the social marginal cost.

Problems of tradable pollution permits:

There are certain problems of pollution permits listed below:

i) It is difficult to determine the quantity of tradable permits. The government can give more or less quantity of permits depending upon their policies.

ii) It is difficult to measure the level of the pollution by the firms. There is a possibility that the firms can hide the true level or shift their production base to some other countries to avoid the surplus cost of the permit. In a globalized economy, the multinational companies often adopt this method to reduce cost.

iii) The administrating cost are likely to incur which cannot be ignored while implementing the scheme.
iv) In case of global pollution permits, countries who emits more than their permissible limit are required to buy it from other countries with excess permit limit. More often it happens with the industrially advanced rich developed countries who buy the permit from the less developed countries. So, the aim to reduce pollution is achieved if such tactics are adopted.
v) There are some schemes such as offsetting schemes. In case of carbon emission beyond permissible limit if a firm agrees to plant trees to offset or compensate the pollution level, then such a permit is to be given to the firm. But once these firms get permit whether they really planted the trees or not is the questionable in many cases.
vi) Many a times political interference defeats the aim of the tradable permits and offers more quantity of permits to the firms.

VI. QUOTAS (COMMAND AND CONTROL INSTRUMENTS)

Quotas or Command and control instruments prescribes certain rules and standards to comply with by using sanctions. This instrument assigns a specific amount of emission to the polluters. So, the polluters are allowed a specific amount of emission which helps to reduce pollution.
In simple words quota means setting up of a limit or permission for a certain amount of a targeted thing. It is often used in Environmental legislation where policy makers intend to impose restrictions to ensure environmental sustainability. Based on a specific formula, these quotas /permits sanctioned to each individual or firm participant. These permits can be distributed by using several options
i) They can be given to the current users / polluter firms
ii) Buying and selling in the market freely determined by the forces of demand and supply
iii) Auctioning off them so that it is to be given to the highest bidder. In each of these cases the permits are transferable in nature.

V. CHOICE BETWEEN TAXES AND QUOTAS

Taxes and Quotas both have their pros and cons:
1. Taxes are the direct payment in monetary terms made by the firms to government for emission of pollutants. Whereas Quotas are the quantitative limits imposed on the emission of pollutants. In case of taxes since there is a direct payment in proportion to the emission, the pinching is more. So, the polluters try to optimize the cost of tax payment by reducing the emission. In this way taxes are effective than the quotas.
2. Measurement of taxes in proportion to the emission is difficult. They are generally measured as equal to the cost of negative externality. To measure taxes are therefore difficult in real world. As against this quota are easy to determine based on a specific formula in real world.
3. Taxes have a direct positive impact on reduction of pollution when due to its heavy payment polluters try to reduce the emission. In case of quotas and permits since they are transferable in nature it does not have any direct impact on reducing the emission of pollutants. To conclude, imposition of only tax or only quotas is not effective. Many a times hybrid policies of both these instruments in required. The choice between taxes and quotas is to be determined on the basis of negative externalities and distribution of cost. The success or the failure of these instruments are mainly depended on the policy implementation and objectives to a large extent.